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Introduction

- ***Circuit Analysis*** is essential to study of Engineering.
- Additionally, it is an education in ***Logical Thinking*****.**
- Application of three simple scientific laws:
  - **Ohm's Law,**
  - **Kirchhoff's Voltage Law (KVL) and**
  - **Kirchhoff's Current Law (KCL).**

- A study of more appropriately ***" Linear Circuit Analysis"*****.**
  - Non-linear models, though much more accurate, are more complex as well.
- Linear circuit analysis can be separated into four broad categories: (1) ***dc analysis***, where the energy sources do not change with time; (2) ***transient analysis***, where things often change quickly; (3) ***sinusoidal analysis***, which applies to both ac power and signals; and (4) ***frequency response,***

Introduction

- We begin our journey with the topic of ***resistive circuits***, which may include simple examples such as ***a flashlight or a toaster***. This provides us with a perfect opportunity to learn a number of very powerful engineering circuit analysis techniques, such as ***nodal analysis***, ***mesh analysis, superposition, source transformation, Thévenin's theorem, Norton's theorem,*** and several methods for simplifying networks of components connected in series or parallel.
- In circuit analysis, we refer to ***transient analysis*** *as* the suite of techniques used to study circuits which are suddenly energized or de-energized.
- Fortunately, for situations where equipment is to run long enough for transient effects to die out, and this is commonly referred to as **ac or sinusoidal analysis, or sometimes *phasor analysis*.**

Introduction

- The final leg of our journey deals with a subject known as ***frequency response***. Working directly with the differential equations obtained in time-domain analysis helps us develop understanding of the operation of circuits containing energy storage elements (e.g., capacitors and inductors). ***Laplace and Fourier analysis***, allow us to transform differential equations into algebraic equations. Such methods also enable us to design circuits to respond in specific ways to particular frequencies.
- *Circuit analysis* teaches us to learn how-:
  - **To be methodical,**
  - **To determine *the goal* of a particular problem,**
  - **To work with the information given to develop *a plan* of action,**
  - **To *implement* the plan to obtain a solution and finally,**
  - **To *verify* that the proposed solution is accurate.**

Introduction….contd

- **In electrical / electronics engineering, there is a need to transfer energy / communicate from one point to another.**
- **This requires an *interconnection* of certain devices.**
- **Such interconnection is called *electrical / electronic circuit.***
- **Each component of the circuit is called an *element*.**
- ***Circuit Analysis* means a study of the behavior of the circuit.**
- This course is focused on developing our ability to analyze and solve problems because it is the starting point in every engineering situation. *The philosophy of this course is that we need clear explanations, well-placed examples, and plenty of practice to develop such an ability*. Therefore, elements of design are integrated into end-of-chapter problems.

Introduction….contd

- We, therefore, need to study basic concepts of :-
  - Charge,
  - Current,
  - Voltage or Potential Difference,
  - Work / Energy
  - Power and
  - Circuit Elements.
- Units and Scales !

Units & Scales

- **The value of any measurable quantity requires both a *number* and a *unit* such as *"3 meters".***
- **Since 1960, General Conference on Weights and Measures adopted the International System of Units (*SI*).**
- ***The 7 basic SI units* of physical quantities, assumed to be mutually independent, are :-**

Quantity	Basic Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Thermodynamic Temp	kelvin	K
Luminous Intensity	candela	cd
Amount of Substance	mole	mol

• ….contd !

Units & Scales

TABLE 1.1

Comparison of the English and metric systems of units.

English	Metric		
	MKS	**CGS**	**SI**
Length: Yard (yd) (0.914 m)	Meter (m) (39.37 in.) (100 cm)	Centimeter (cm) (2.54 cm = 1 in.)	Meter (m)
Mass: Slug (14.6 kg)	Kilogram (kg) (1000 g)	Gram (g)	Kilogram (kg)
Force: Pound (lb) (4.45 N)	Newton (N) (100,000 dynes)	Dyne	Newton (N)
Temperature: Fahrenheit (°F) $\left(= \frac{9}{5}°C + 32\right)$	Celsius or Centigrade (°C) $\left(= \frac{5}{9}(°F - 32)\right)$	Centigrade (°C)	Kelvin (K) K = 273.15 + °C
Energy: Foot-pound (ft-lb) (1.356 joules)	Newton-meter (N•m) or joule (J) (0.7376 ft-lb)	Dyne-centimeter or erg (1 joule = 10^7 ergs)	Joule (J)
Time: Second (s)	Second (s)	Second (s)	Second (s)

- ….contd!

Units & Scales ….contd

- ***All else is derived from these basic 7 units.***
- **Fundamental *unit of work* (or energy) is *joule*.**
- ***One joule* of work is done when a force of *one newton* moves an object through *one meter*.**
- ***One joule* in SI base units is *kg m^2 s^{-2}*.**
- ***One Btu* is 1055 joules.** (Btu is the amount of work needed to raise the temperature of one pound of water by one degree Fahrenheit.)
- ***Power* is defined as the rate at which work is done or energy is expended.**
- **The fundamental *unit of power* is the watt.**
- ***One watt* is defined as *1 joule/sec*.**
- **One watt is equal to 1/746 horsepower (hp).**
- **A power level of 1 hp is approximately equivalent to 746 watt s (W) or 0.746 kilowatt s (kW).**
- **One kilowatthour (kWh) is 3.6 X 10^6 J.**

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

- **The Charge.**

Charge

- All matter consists of atomic particles.
- *Charge* is an electrical property of the atomic particles.
- Unit of measure of charge is *coulomb 'C'* (Coulomb 1736 – 1806).
- Charge *'e'* of an electron is *negative* & equal in magnitude to 1.602×10^{-19} C. Concurrently, a single proton has a charge of $+ 1.602 \times 10^{-19}$ C.
- One coulomb of charge is present on $1 / 1.602 \times 10^{-19}$ electrons. This is equal to 6.24×10^{18} electrons.
- Charge is represented by 'Q' (time-invariant) and q(t) or 'q' when it varies with time.
- Law of conservation of charge states that charge can neither be created nor destroyed.

Current

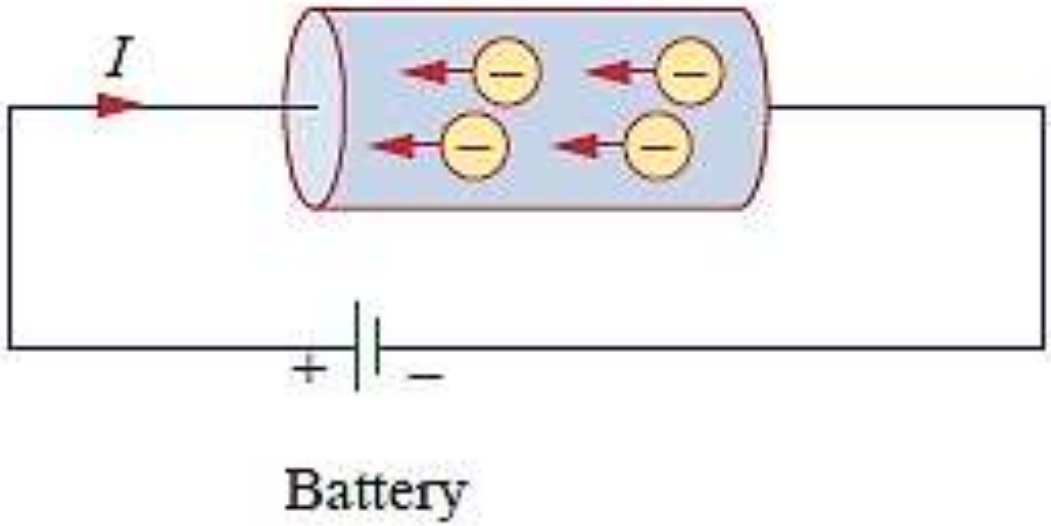
- **A unique feature of charge is that it is mobile.**
- **Charge in motion is called electric current represented by 'I' or i(t) or 'i'.**
- **When force is applied or work is done, negative charges can be made to move in one direction while the positive charges can be made to move in the opposite direction, thus constituting *a current*.**
- **It is *conventional* to take current flow as movement of *positive charges* (Benjamin Franklin 1706 – 1790).**
- **A convention is a standard way of describing something so that others in the profession can understand.**
- **Mathematically current 'i' and charge 'q' are related by :-**

$i = dq / dt$ where current 'i' is measured in amperes (A)

So *1 ampere = 1 coulomb / second*

Current

- We now consider the flow of electric charges. A unique feature of electric charge or electricity is the fact that it is ***mobile***; that is, it can be transferred from one place to another, where it can be converted to another form of energy.



Current

- **The current present in a discrete path, such as a metallic wire, has both a *numerical value* and a *direction* associated with it; it is a measure of the rate at which charge is moving past a given reference point in a specified direction.**
- **Let q(t) be the total charge that has passed the reference point since an arbitrary time *t = 0*, moving in the defined direction.**
- **As an example, Fig. 2.2 shows a history of the total charge *q(t)* that has passed a given reference point in a wire**

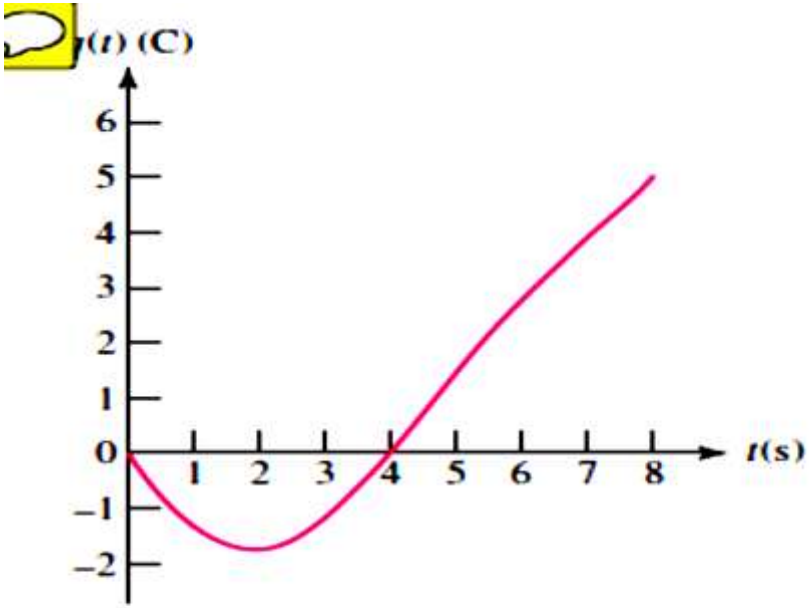


FIGURE 2.2 A graph of the instantaneous value of the total charge $q(t)$ that has passed a given reference point since $t = 0$.

$$i = \frac{dq}{dt} \qquad [1]$$

Current

- **Using Eq. [1], we compute the instantaneous current and obtain Fig. 2.3. The charge transferred between time t_0 and t may be expressed as a definite integral:**

$$\int_{q(t_0)}^{q(t)} dq = \int_{t_0}^{t} i\,dt'$$

The total charge transferred over all time is thus given by

$$q(t) = \int_{t_0}^{t} i\,dt' + q(t_0) \qquad [2]$$

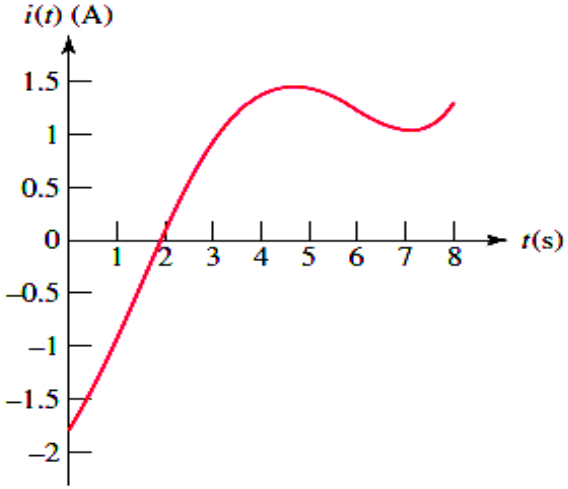
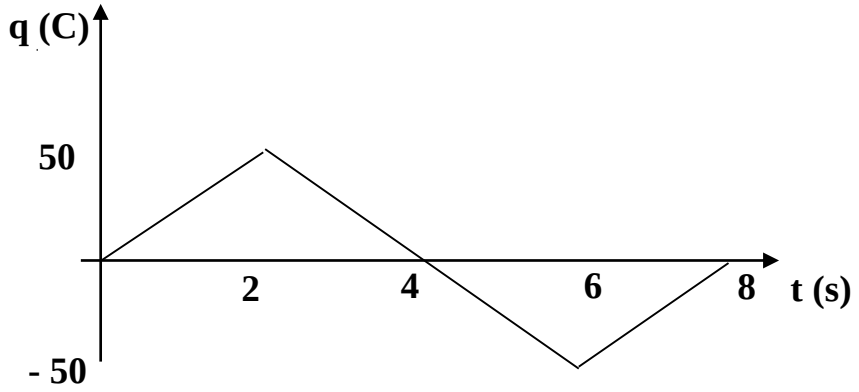


FIGURE 2.3 The instantaneous current $i = dq/dt$, where q is given in Fig. 2.2.

Example : Charge & Current

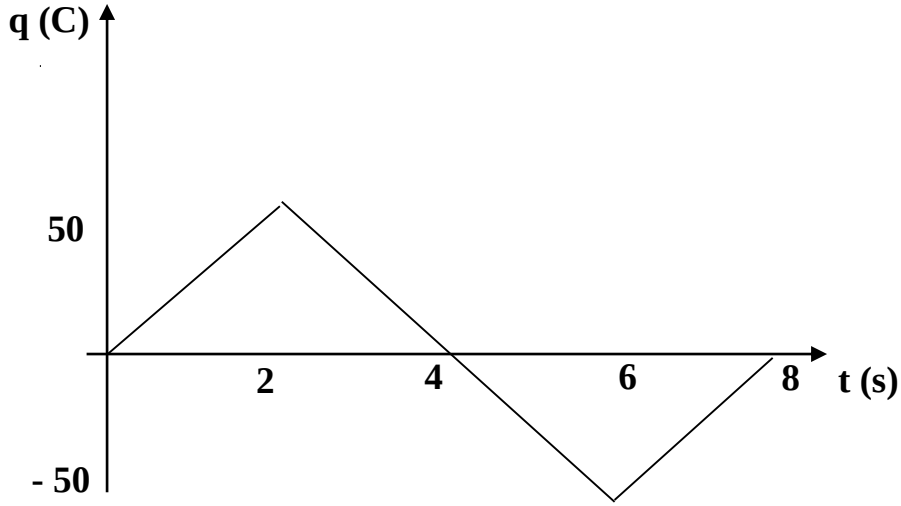
- **The charge flowing in a wire is plotted as follows; sketch the corresponding current.**

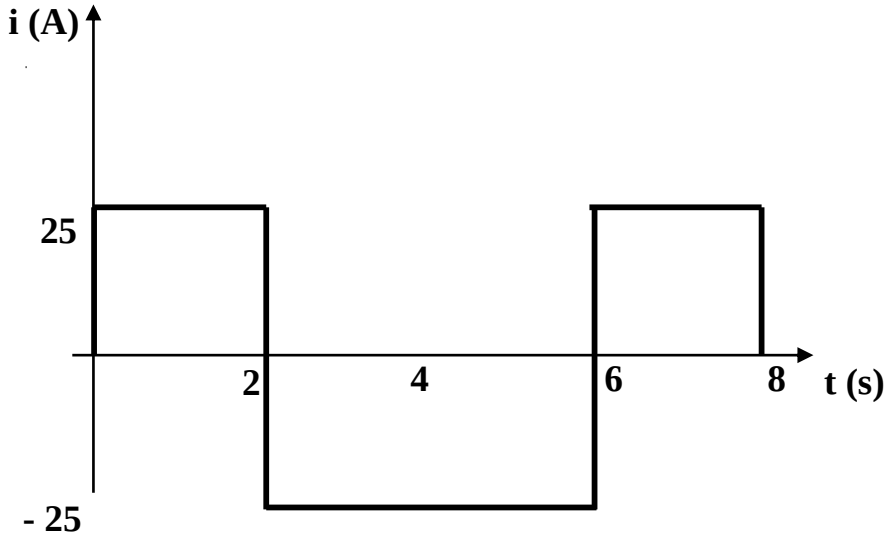


- **Notes: At t = 1 sec the charge q(1) is measured as 25 coulomb. So between 0 sec and 1 sec the charge flow is 25 coulomb in 1 sec; hence current i(1) = 25 A.**
  - **Again at t = 7 sec the charge is q(7) = - 25 A. From q(6) of – 50 A the change is final value – initial value that is – 25 – (- 50) = 25 so i(7) = 25 A.**

- **…contd.**

Example : Charge & Current ….contd





Current

- **We create a graphical symbol for current by placing an arrow next to the conductor.**
- ***In Fig. 2.5a,*** **the direction of the arrow and the value 3A indicate either that *a net positive charge of 3 C/s is moving to the right* or that *a net negative charge of −3 C/s is moving to the left* each second.**
- ***In Fig. 2.5b,*** **there are again two possibilities: either −3A is flowing to the left or +3A is flowing to the right.**

FIGURE 2.5 Two methods of representation for the exact same current.

- **<u>A nonelectrical analogy</u> : e.g., a deposit can be viewed as either a *negative cash flow out of your account* or *a positive flow into your account.***

- **It is essential that we realize that the current arrow does not indicate the *"actual"* direction of current flow but is simply part of *a convention* that allows us to talk about "the current in the wire" in an unambiguous manner.**
- **The arrow is a *fundamental part* of the definition of a current! Thus, to talk about the value of a current $i_1(t)$ without specifying the arrow is to discuss an undefined entity. For example, Fig. 2.6 *a* and *b* are meaningless representations of $i_1(t)$, whereas *Fig. 2.6c is complete*.**

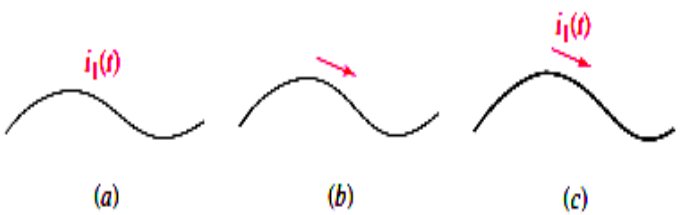
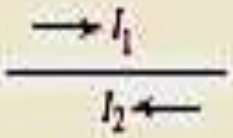


FIGURE 2.6 (*a*, *b*) Incomplete, improper, and incorrect definitions of a current. (*c*) The correct definition of $i_1(t)$.

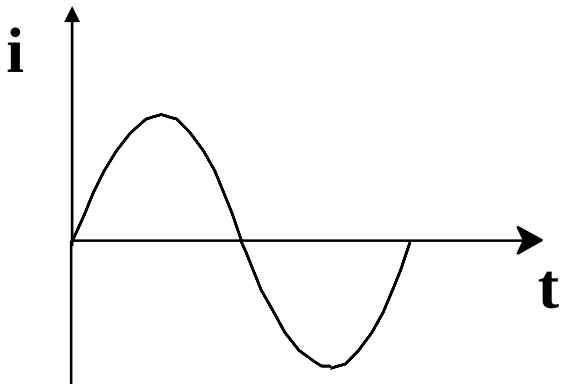
PRACTICE

2.4 In the wire of Fig. 2.7, electrons are moving *left* to *right* to create a current of 1 mA. Determine I_1 and I_2.



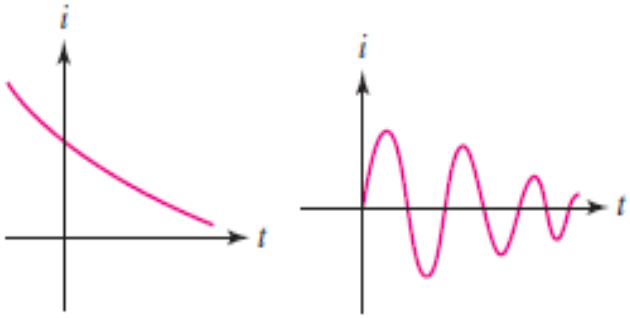
AC & DC Current

- A *unidirectional* current is called *dc or direct current*.
- *Alternating current (ac)* is current that *changes direction cyclically.*



- *Exponential and damped* sinusoid currents!!

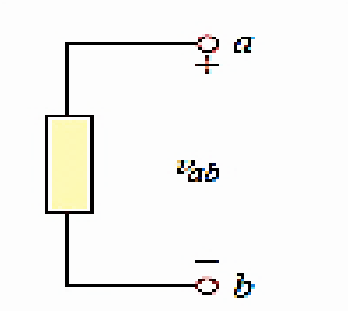
- Voltage !



Voltage

- To move the electron in a conductor in a particular direction requires some *work or energy transfer*.
- This work is performed by an external *electromotive force* (emf), typically represented by the ***battery***.
- This emf is also known as ***voltage or potential difference***. The voltage v_{ab} between two points *a* and *b* in an electric circuit is the energy (or work) needed to move a unit charge from *a* to *b*; mathematically,
- **Mathematically, the relationship is :-**

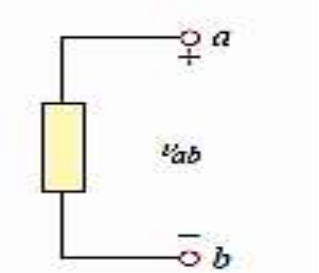
$v_{ab} = dw / dq$ **where 'w' is energy in joules (J) & 'q' is charge in coulombs.**



So **1 volt = 1 joule / coulomb**

Voltage

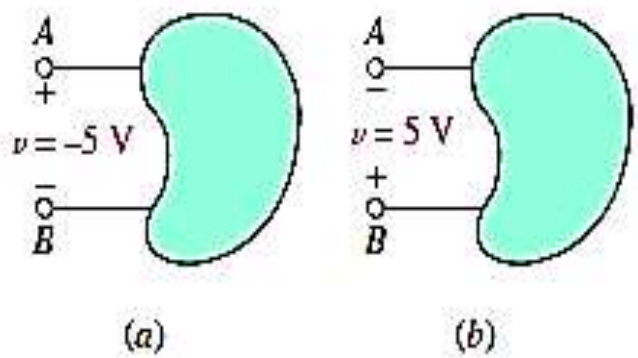
- The v_{ab} can be interpreted in two ways:
- (1) ***point a*** is at a potential of v_{ab} volts higher than ***point b***, or
- (2) the potential at **point *a*** with respect to **point *b*** is v_{ab}



It follows logically that in general

$$V_{ab} = - V_{ba}$$

- **In Fig. 2.9*a*, for example, the placement of the + sign at terminal *A* indicates that *terminal A* is *v volts +ve* with respect to terminal *B*. If we later find that *v* happens to have a numerical value of −5 V, then we may say either that *A* is −5 V positive with respect to *B* or that *B* is +5 V positive with respect to *A*. Other cases are shown in Fig. 2.9*b*, *c*, and *d*.**



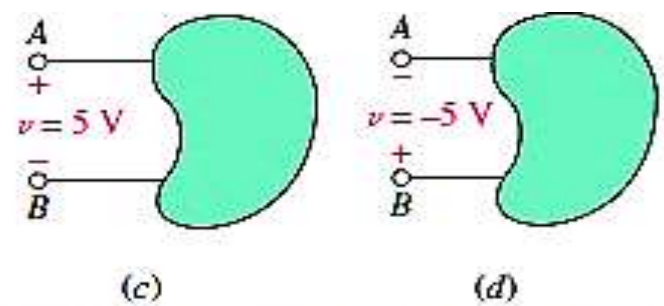


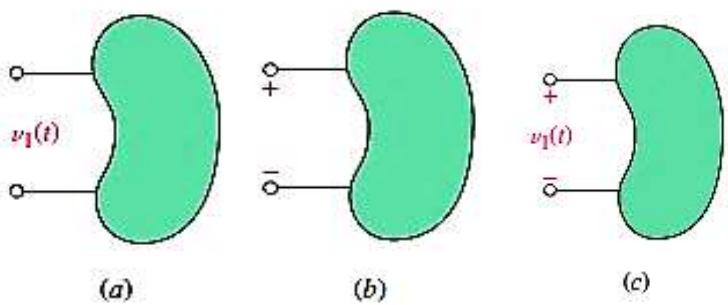
FIGURE 2.9 (*a*, *b*) Terminal *B* is 5 V positive with respect to terminal *A*; (*c*, *d*) terminal *A* is 5 V positive with respect to terminal *B*.

- **…. contd.**

Voltage…contd

- It is essential to realize that the ***plus-minus pair of algebraic signs*** does not indicate the ***"actual"*** polarity of the voltage but is simply part of ***a convention*** that enables us to talk unambiguously about "the voltage across the terminal pair."

- *The definition of any voltage must include a* ***plus-minus sign*** *pair* Using a quantity $v_1(t)$ without specifying the location of the plus-minus sign pair is using an undefined term. Figure 2.10*a* and *b* do *not* serve as definitions of $v_1(t)$; Fig. 2.10*c* does.



PRACTICE

2.5 For the element in Fig. 2.11, $v_1 = 17$ V. Determine v_2.

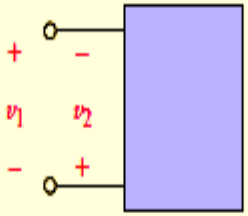


FIGURE 2.11

- **Any question !**

He who *asks a question* is a *fool for* five minutes; he who does not *ask a question* remains a *fool* forever.

(Chinese proverb)

Power

- Current and voltage are the two basic variables in an electric circuit, they are not sufficient by themselves. For practical purposes, we need to know how much ***power*** an electric device can handle. We all know from experience that a ***100-watt bulb*** gives more light than ***a 60-watt bulb***. We also know that when we pay our bills to the electric utility companies, we are paying for the electric ***energy*** consumed over a certain period of time. Thus, power and energy calculations are important in circuit analysis.
- ***Power is defined as the rate at which work is done or energy is expended.***
- ***It is expressed in watts (W). Power is one watt if one joule of energy is expended in transferring one coulomb of charge through the device in one second.***

- **Mathematically, the relationship is :-** $p = dw / dt$

  where 'p' is power in watts (W), 'w' is energy in joules (J) & 't' is time in seconds.

- **Also $p = dw / dt = dw / dq \times dq / dt = vi$**

Power

- **The absorbed power must be proportional both to the number of coulombs transferred per second *(current)* and to the energy needed to transfer one coulomb through the element *(voltage).***
- **The conventions for current, voltage, and power are shown in Fig. 2.12.**

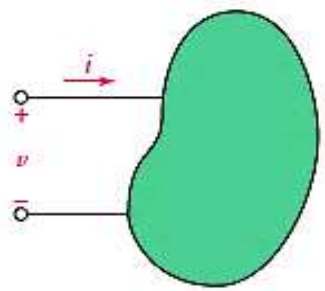
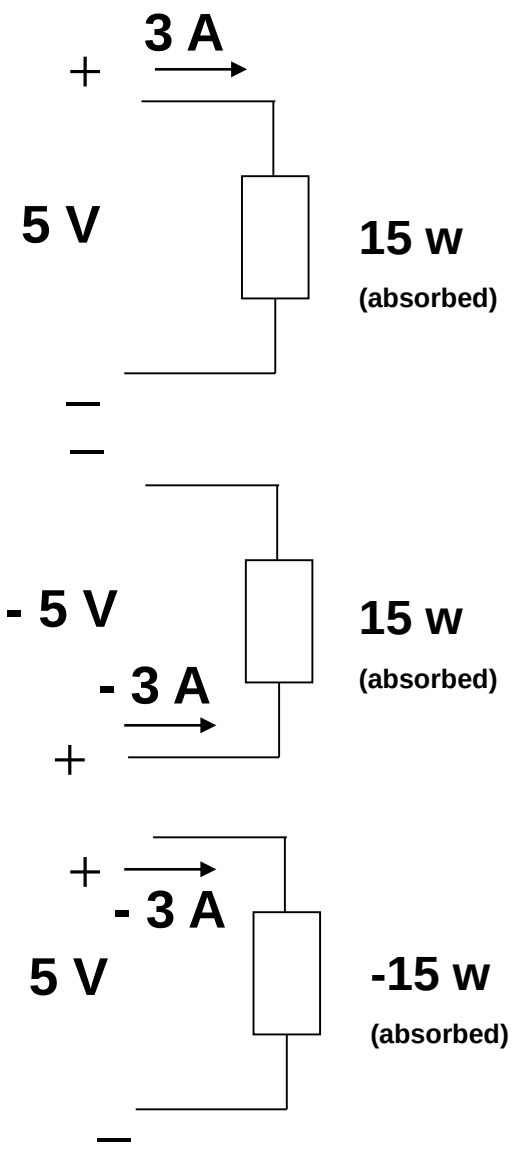


FIGURE 2.12 The power absorbed by the element is given by the product $p = vi$. Alternatively, we can say that the element generates or supplies a power $-vi$.

- Thus, the power ***absorbed or supplied*** by an element is the product of the ***voltage*** across the element and the ***current*** through it. If the power has a **+ve** sign, power is being delivered to or ***absorbed*** by the element. If, on the other hand, the power has a **-ve** sign, power is being ***supplied*** by the element.

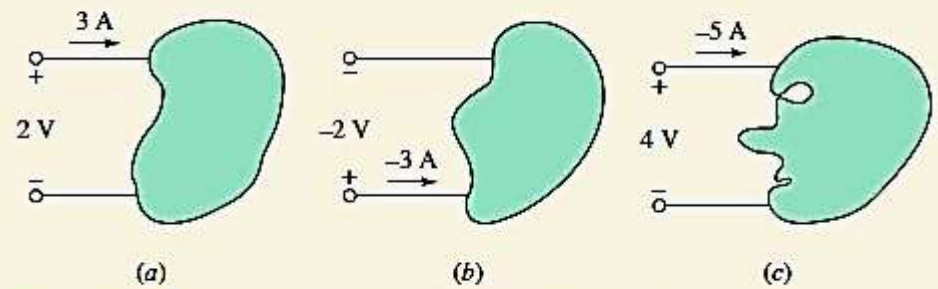
Passive Sign Convention

- The passive sign convention is satisfied when the current arrow is directed into the element at the *plus-marked terminal*.
- It says that if the current arrow and the voltage polarity signs are placed such that the current enters that end of the element marked with the positive sign, then the power *absorbed* can be expressed by the product of the specified current and voltage variables.
  - If the numerical value of the *product is negative, then we say that the element is absorbing negative power*, or that it is actually *generating power* and delivering it to some external element.
    - In the first case the +ve terminal is at 5 volts wrt the other terminal.
    - In the second case the +ve terminal is at – 5 volts wrt the other terminal.



Example : Power

Compute the power absorbed by each part in Fig. 2.13.



■ **FIGURE 2.13** (*a, b, c*) Three examples of two-terminal elements.

In Fig. 2.13*a*, we see that the reference current is defined consistent with the passive sign convention, which assumes that the element is absorbing power. With +3 A flowing into the positive reference terminal, we compute

$$P = (2\text{ V})(3\text{ A}) = 6\text{ W}$$

of power absorbed by the element.

Figure 2.13*b* shows a slightly different picture. Now, we have a current of −3 A flowing into the positive reference terminal. This gives us an absorbed power

$$P = (-2\text{ V})(-3\text{ A}) = 6\text{ W}$$

Thus, we see that the two cases are actually equivalent: A current of +3 A flowing into the top terminal is the same as a current of +3 A flowing out of the bottom terminal, or, equivalently, a current of −3 A flowing into the bottom terminal.

Referring to Fig. 2.13*c*, we again apply the passive sign convention rules and compute an absorbed power

$$P = (4\text{ V})(-5\text{ A}) = -20\text{ W}$$

Since we computed a negative *absorbed* power, this tells us that the element in Fig. 2.13*c* is actually *supplying* +20 W (i.e., it's a source of energy).

PRACTICE : Power

PRACTICE

2.6 Determine the power being absorbed by the circuit element in Fig. 2.14*a*.

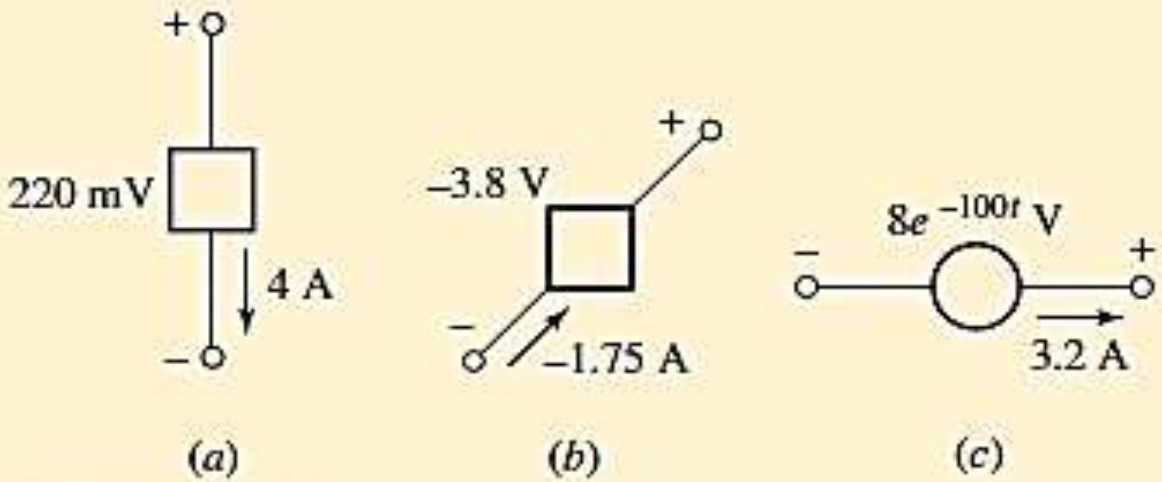


FIGURE 2.14

2.7 Determine the power being generated by the circuit element in Fig. 2.14*b*.

2.8 Determine the power being delivered to the circuit element in Fig. 2.14*c* at $t = 5$ ms.

Example : Power

- How much energy does a 100 - watt electric bulb consume in two hours?

We know p = dw / dt where w is in joules and t in seconds.

- So 100 = w (joules) / (2X60X60)
- Hence w = 720 k joules.

Also pdt = dw

- So watt X sec = joules
- Or watt X 3600 X sec = joules X 3600
- Hence 1watt-hour = 3600 joules
- And 200 watt - hr = 720 k joules *(watt-hr is the unit used by electrical companies)*

The Circuit Elements !

Thanks